

OBSTRUCTION RESIDUES FOR THE $(AN + B)$ FAMILY: BIAS CONJUGATION AND MULTIPLIER ORDER

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ABSTRACT. For odd integers a, b with $|a| \geq 3$ and $b \neq 0$, the map $T_{a,b}(n) := (an + b)/2^{v_2(an+b)}$ acts on the odd integers. The *obstruction residues* $\mathcal{O}_L^{(a,b)} \subseteq \{1, \dots, 2^L\}$ are the odd residues at level L for which an explicit two-track parallel reduction terminates with a vanishing affine certificate $\Phi_J = 0$; the T_- -manuscript proves a constructive lift, positive density c_W , and full factor complexity for the special case $(a, b) = (3, -1)$.

We extend the theory to the whole class via two bridges. Along the bias axis, the multiplicative conjugation $\Psi_b(r) := br \bmod 2^L$ is a shift-preserving bijection $\mathcal{O}_L^{(a,1)} \rightarrow \mathcal{O}_L^{(a,b)}$, so the cardinality $|\mathcal{O}_L^{(a,b)}|$ and the asymptotic density $c_W^{(a)}$ depend only on a . Along the multiplier axis, a diophantine identity for the reduction gives, modulo $|a|$, the order lemma $\delta_J \equiv 0 \pmod{e}$ with $e := \text{ord}_{|a|}(2)$ (the T_- -manuscript's parity lemma is the case $a = 3$). The density $c_W^{(a)}$ exists as a monotone limit, and existence of obstructions ($c_W^{(a)} > 0$) is equivalent to a finite diophantine solvability condition; a 2-adic Haar interpretation places $c_W^{(a)}$ in the Bernstein–Lagarias tradition. We further prove $L_{\min}(a) \geq e + 1$ (outside the negative-Mersenne case), a diophantine criterion for the levels carrying $J = 1$ atoms (with the Mersenne multipliers as its principal solutions), the no-shift-zero-atom theorem for all $|a| \geq 3$ outside the negative-Mersenne case, and a polynomial bound on fixed- J atom counts with a constructive extremal family at $J = L - 4$ for $a = 3$. Beyond these theorems the quantitative size of $c_W^{(a)}$ is so far only empirical: across the tested multipliers $c_W^{(a)} > 0$ and $c_W^{(a)} \approx C(a) 4^{-e}$ for an a -dependent factor $C(a)$ that stays bounded in the tested range; whether $C(a)$ is bounded in general (equivalently $c_W^{(a)} = \Theta(4^{-e})$) and whether $\alpha(a) < 2$ are the principal conjectures and remain open (a Wirsching-style operator is sketched as one route). The boundary $|a| = 1$ is excluded — $\mathcal{O}_L^{(1,b)} = \emptyset$ for all L at $a = +1$, while $a = -1$ is degenerate — confirming the necessity of $|a| \geq 3$. The negative-Mersenne multipliers $a = -(2^{v_M} - 1)$ admit a finer sub-class hierarchy treated in a separate manuscript.

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1. INTRODUCTION

1.1. Setting. For odd integers a, b with $|a| \geq 3$ and $b \neq 0$, the *affine accelerated map* $T_{a,b}$ acts on the set $\mathbb{N}_{\text{odd}}$ of positive odd integers by

$$T_{a,b}(n) := \frac{an + b}{2^{v_2(an+b)}}.$$

Two specialisations are classical: $(a, b) = (3, 1)$ is the Collatz map T_+ ([4, 3]), and $(a, b) = (3, -1)$ is the $3n - 1$ map T_- , which is known to admit at least three distinct cycles (the fixed cycle through 1, the $\{5, 7\}$ -cycle, and the 7-cycle through 17; see [3]). The Collatz conjecture asserts that every T_+ -trajectory reaches 1; the strongest unconditional progress towards it is Tao’s theorem that almost all orbits attain almost bounded values [9]. The analogous question for T_- asks whether further cycles exist beyond the three known ones. Generalised $(an + b)$ maps go back to Hasse’s generalised Syracuse algorithm, whose theory is developed by Matthews and Watts [5] and Möller [6] (the latter’s title is literally “Hasse’s generalisation of the Syracuse algorithm”). The broader class of piecewise-defined generalised Collatz iterations is computationally universal — Conway’s undecidability of such iteration problems [2] — which is part of what makes an exact, level-by-level algebraic characterisation of a structured subfamily worthwhile; the present obstruction-residue viewpoint is exactly such a characterisation, rather than a measure-theoretic or heuristic one. The 2-adic conjugation that underlies our bias-axis bridge (Section 3) and the Haar interpretation of Section 4 are in the tradition of the $3x + 1$ conjugacy map of Bernstein and Lagarias [1], who realise the Collatz dynamics as a conjugate of the shift on $\mathbb{Z}_2$.

Both maps share an algebraic feature that turns out to be visible at the level of *residues modulo* 2^L . The T_- -manuscript [8] introduces a family of residue classes $\mathcal{O}_L \subseteq \{1, \dots, 2^L\}$ characterised by an algebraic criterion on a two-track parallel reduction, and proves three structural results for this family:

- (1) a constructive lift between modular levels L and $L + 1$;
- (2) a strictly positive asymptotic density $c_W \in (0, \frac{1}{2}]$ (only odd residues can be obstructions, so $\frac{1}{2}$ is the trivial ceiling) with the rigorous lower bound $c_W \geq 7556/65536 \approx 0.115$;
- (3) full factor complexity $p_W(\ell) = 2^\ell$ of the associated language of binary representations.

All three rest on a *parity lemma*: at termination of the parallel reduction, the difference $\delta_J = V_K^{(J)} - V_I^{(J)} - v$ of cumulative valuations is forced to be an even integer.

1.2. The two bridges. The present paper develops the same theory in the wider $(an + b)$ setting, $|a| \geq 3$ and b odd, by isolating two bridges from the T_- -manuscript’s special case.

Bridge 1: bias axis (multiplicative conjugation). The multiplicative map $\Psi_b(r) := br \bmod 2^L$ is a shift-preserving bijection $\mathcal{O}_L^{(a,1)} \rightarrow \mathcal{O}_L^{(a,b)}$ for every odd $b \neq 0$ (Theorem 3.1). The T_- -manuscript's involution $r \mapsto 2^L - r$ that transfers results from T_- to T_+ ([8, Section 7]) is the special case Ψ_{-1} . As a consequence, the cardinalities $|\mathcal{O}_L^{(a,b)}|$ and the asymptotic density $c_W^{(a,b)}$ depend only on a , not on b (Corollary 3.3). We write $c_W^{(a)} := c_W^{(a,b)}$ unambiguously.

Bridge 2: multiplier axis (order lemma). Reducing the *diophantine identity* for $\Phi_J = 0$ modulo $|a|$ forces $\delta_J \equiv 0 \pmod{e}$, where $e := \text{ord}_{|a|}(2)$ (Lemma 3.6). The T_- -manuscript's parity lemma is the case $a = 3, e = 2$. Unlike the conjugation theorem, the order lemma restricts but does not preserve structure across a : the step size e of the shift index grows with the multiplicative order and drives an empirical density law $c_W^{(a)} \approx C(a) 4^{-e}$ across the tested range $|a| \in \{3, 5, 7, 9, 15, 17, 21, 31\}$ (Section 8).

1.3. Main results. For odd a, b with $|a| \geq 3$ and $b \neq 0$ and a level $L \geq 1$, the obstruction set $\mathcal{O}_L^{(a,b)}$ at level L is defined in Section 2 (Definition 2.2) as the set of odd residues r for which the two-track parallel reduction terminates with $\Phi_J = 0$. The shift index $\delta_J = \delta_J(r, L)$ is the difference of cumulative valuations at termination; the constant $e := \text{ord}_{|a|}(2)$ is the multiplicative order of 2 modulo $|a|$.

Theorem 1.1 (Conjugation, b-axis universality). *For every odd $b \neq 0$ the multiplicative map $\Psi_b(r) := br \bmod 2^L$ is a shift-preserving bijection $\mathcal{O}_L^{(a,1)} \rightarrow \mathcal{O}_L^{(a,b)}$. In particular, the cardinality $|\mathcal{O}_L^{(a,b)}|$ and the asymptotic density $c_W^{(a,b)} = \lim_L |\mathcal{O}_L^{(a,b)}|/2^L$ depend only on a .*

Theorem 1.2 (Diophantine identity, order lemma). *For every $r \in \mathcal{O}_L^{(a,b)}$ with termination index J ,*

$$a^{J+1} = \sum_{j=0}^J a^{J-j} X_j, \quad X_j := 2^{V_K^{(j)}} - 2^{V_I^{(j)}+v}.$$

Reducing modulo $|a|$ gives $\delta_J \equiv 0 \pmod{e}$.

Theorem 1.3 (Density limit and existence criterion). *The density $c_W^{(a,b)} = \lim_L |\mathcal{O}_L^{(a,b)}|/2^L$ exists as a non-decreasing limit, and $c_W^{(a)} > 0$ if and only if the diophantine identity above is realisable for at least one finite valuation tuple $(J, V_K^{(j)}, V_I^{(j)}, v)$.*

A measure-theoretic reading of this density — $c_W^{(a)} = \mu_2(\mathcal{O}^{2\text{-ad},(a,b)})$, the Haar measure of the open set $\mathcal{O}^{2\text{-ad},(a,b)} \subseteq \mathbb{Z}_2^\times$ of 2-adic units that are eventually obstructions — is recorded as Remark 4.4, in the Bernstein–Lagarias tradition.

Theorem 1.4 (No shift-zero atom, generalised). *Assume a is not a negative Mersenne number $-(2^{v_M} - 1)$. For every level L ,*

$$|\mathcal{A}_L^{(a,b),G_0}| = 0;$$

every atomic obstruction has nonzero shift index.

Proposition 1.5 (Atom localisation at fixed J). *At fixed termination index J the atomic $G_{\neq 0}$ -count is bounded by $|\mathcal{A}_L^{(a,b),G_{\neq 0}}|_J \leq \binom{L-1}{J} = O_J(L^J)$ (Proposition 6.3): every such atom drives one track to the boundary of its modulus, and the boundary valuation data then determine the residue uniquely. For $(a,b) = (3, -1)$ the residue $r := 2^{L-2} + 3$ is an atomic obstruction at every even level $L \geq 6$ with $J = L - 4$ and shift index 1, realising the empirical extreme $J_{\max} = L - 4$.*

Theorem 1.6 (Degenerate edge case). *For $a = +1$ and every odd b , $\mathcal{O}_L^{(a,b)} = \emptyset$ for every L . For $a = -1$ the map degenerates ($an + b$ vanishes on part of the domain) and direct enumeration gives $|\mathcal{O}_L^{(a,b)}| = 2^{L-2}$ (Remark 9.2); neither value of $|a| = 1$ supports the genuine obstruction dynamics that require $|a| \geq 3$.*

Theorems 1.1–1.4, Proposition 1.5 and Theorem 1.6 are proven across Sections 3 (conjugation and order lemma), 4 (density limit and existence criterion), 6 (no-shift-zero atom and atom localisation), and 9 (degenerate edge case). Two further results (Theorem 5.1 on the lower bound for the smallest non-trivial level and Proposition 5.2 on the characterisation of $J = 1$ atoms, for positive a) are stated in Section 5. The principal open problems — $\alpha(a) < 2$ and the existence conjecture $c_W^{(a)} > 0$ for non-Mersenne $|a|$ — are discussed in Section 7, and the empirical density scaling $c_W^{(a)} \approx C(a) 4^{-e}$ (with the boundedness of $C(a)$ itself conjectural) is reported in Section 8.

1.4. Relation to the negative-Mersenne case. For $a = -(2^{v_M} - 1)$ Mersenne with $v_M \geq 2$, the $(an + b)$ theory admits a finer sub-class hierarchy indexed by the shift index $j \in \mathbb{Z}$ (the iterated A -, B -, C -, and F_1 -families, plus a $j = -1$ family G_1 specific to negative a); the T_- -manuscript-style density bound becomes $c_W^{(a)} \geq 1/(2|a|)$, asymptotically sharp in v_M . We treat this case in a separate manuscript (in preparation) and only cite the relationship where it sharpens the general theory.

1.5. Outline. Section 2 fixes notation and defines the two-track parallel reduction. Section 3 proves the four foundational tools (conjugation, order lemma, diophantine identity, and the series representation of $c_W^{(a)}$). Section 4 establishes the density limit and reduces existence of obstructions to a finite diophantine solvability condition (with a 2-adic Haar interpretation). Section 5 establishes the lower bound $L_{\min}(a) \geq e + 1$ and characterises the levels at which $J = 1$ atoms exist. Section 6 contains the structural results over a : the generalised no-shift-zero-atom theorem, polynomial atom bounds at fixed J , and the constructive $L - 4$ extremal family. Section 7 discusses the

principal open problem $\alpha(a) < 2$ (with a Wirsching-style operator sketched as one route). Section 8 reports the empirical scaling $c_W^{(a)} = \Theta(4^{-e})$. Section 9 treats the degenerate boundary $|a| = 1$ ($a = +1$ empty, $a = -1$ degenerate). Section 10 records the relationship to the companion manuscript's negative-Mersenne sub-class analysis. Appendix A lists two refuted hypotheses that motivated the present development.

2. SETUP: PARALLEL REDUCTION AND OBSTRUCTION RESIDUES

2.1. The two-track parallel reduction. Fix odd integers a, b with $|a| \geq 3$ and $b \neq 0$. Let r be a positive odd integer, and write

$$v := v_2(r + b), \quad m := (r + b)/2^v.$$

Since b is odd (which we always assume) and r is odd, the sum $r + b$ is even, so $v = v_2(r + b) \geq 1$ and $m = (r + b)/2^v$ is odd.

Fix a level $L > v$. The parallel reduction proceeds simultaneously on two tracks:

- the *K-track*, $(a_K^{(0)}, b_K^{(0)}) := (r, 2^L)$;
- the *I-track*, $(a_I^{(0)}, b_I^{(0)}) := (m, 2^{L-v})$.

At each step $j \geq 0$, let $A_K^{(j)} := a \cdot a_K^{(j)} + b$, $A_I^{(j)} := a \cdot a_I^{(j)} + b$, $v_K^{(j)} := v_2(A_K^{(j)})$, $v_I^{(j)} := v_2(A_I^{(j)})$, and update

$$a_K^{(j+1)} := A_K^{(j)} / 2^{v_K^{(j)}}, \quad b_K^{(j+1)} := a \cdot b_K^{(j)} / 2^{v_K^{(j)}},$$

provided $v_K^{(j)} < v_2(b_K^{(j)})$; otherwise the K-track terminates at index j . The I-track is updated analogously with $v_I^{(j)}$ and $b_I^{(j)}$. Cumulative valuations are $V_K^{(j)} := \sum_{i < j} v_K^{(i)}$ and $V_I^{(j)} := \sum_{i < j} v_I^{(i)}$, so that the residual valuations of the moduli are $L - V_K^{(j)}$ and $L - v - V_I^{(j)}$ respectively.

The reduction proceeds as long as both tracks can step; it terminates as soon as either condition fails. We write $J = J(r, L)$ for the index of termination.

2.2. The affine certificate.

Lemma 2.1. *For each j with $0 \leq j \leq J$, there exist $c_j, d_j \in \mathbb{Q}$ with*

$$a_I^{(j)} = c_j \cdot a_K^{(j)} + d_j \quad \text{in } \mathbb{Q},$$

where $c_0 = 2^{-v}$, $d_0 = b \cdot 2^{-v}$, and

$$c_{j+1} = c_j \cdot 2^{v_K^{(j)} - v_I^{(j)}}, \quad d_{j+1} = \frac{a \cdot d_j + b(1 - c_j)}{2^{v_I^{(j)}}}.$$

Proof. At $j = 0$: $m = (r + b)/2^v = 2^{-v}r + b \cdot 2^{-v}$. Inductively, $A_I^{(j)} = a(c_j a_K^{(j)} + d_j) + b = c_j A_K^{(j)} + (ad_j + b(1 - c_j))$. Dividing by $2^{v_I^{(j)}}$ and using

$$A_K^{(j)} = 2^{v_K^{(j)}} a_K^{(j+1)}.$$

$$a_I^{(j+1)} = c_j \cdot 2^{v_K^{(j)} - v_I^{(j)}} \cdot a_K^{(j+1)} + \frac{ad_j + b(1 - c_j)}{2^{v_I^{(j)}}}. \quad \square$$

Write $\delta_j := V_K^{(j)} - V_I^{(j)} - v$. From the update rule, $c_j = 2^{\delta_j}$ for every j . Define the *affine certificate* at termination as

$$\Phi_J := b(1 - c_J) + a \cdot d_J.$$

Definition 2.2 (Obstruction residue). Let r be a positive odd integer with $r + b$ even, $v := v_2(r + b) \geq 1$, and $L > v$. We call r an *obstruction residue* for $T_{a,b}$ at level L if $\Phi_J(r, L) = 0$. The set of all such residues in $\{1, \dots, 2^L\}$ is denoted $\mathcal{O}_L^{(a,b)}$.

Remark 2.3 (Terminal synchronisation). The defining condition $\Phi_J = 0$ is equivalent to an *exact* synchronisation of the two tracks at termination. From the affine relation $a_I^{(j)} = c_j a_K^{(j)} + d_j$ and $A_K^{(j)} = a a_K^{(j)} + b$ one has the identity

$$A_I^{(j)} = a a_I^{(j)} + b = c_j A_K^{(j)} + \Phi_j, \quad \Phi_j := b(1 - c_j) + a d_j,$$

so $\Phi_J = 0 \iff A_I^{(J)} = c_J A_K^{(J)}$. Since $v_2(A_K^{(J)}) = v_K^{(J)}$ and $c_J = 2^{\delta_J}$, this forces $v_I^{(J)} = v_K^{(J)} + \delta_J$; substituting this and $V_I^{(J)} = V_K^{(J)} - v - \delta_J$ into the two stop conditions $v_K^{(J)} \geq L - V_K^{(J)}$ and $v_I^{(J)} \geq L - v - V_I^{(J)}$ makes them identical, so *both tracks meet the termination condition at the same index J* . We call the terminal step *exact* when $v_K^{(J)} = L - V_K^{(J)}$ (equivalently the I-track equality holds) and *strict* when $v_K^{(J)} > L - V_K^{(J)}$; by the synchronisation the two tracks share this type. The certificate is thus the algebraic shadow of a synchronisation of stopping times for $T_{a,b}$ -iterates of r and m ; the present paper develops only the algebraic side.

2.3. The shift index. The integer $\delta_J(r, L) = V_K^{(J)} - V_I^{(J)} - v$ controls the fine structure of $\mathcal{O}_L^{(a,b)}$. We refer to it as the *shift index*. The shift-zero set is

$$\mathcal{O}_L^{(a,b),G_0} := \{r \in \mathcal{O}_L^{(a,b)} : \delta_J(r, L) = 0\}, \quad \mathcal{O}_L^{(a,b),G \neq 0} := \mathcal{O}_L^{(a,b)} \setminus \mathcal{O}_L^{(a,b),G_0},$$

and analogously for $\mathcal{A}_L^{(a,b),G_0}, \mathcal{A}_L^{(a,b),G \neq 0}$ once atomic obstructions are defined below.

2.4. Lifts and atoms. The obstruction sets at consecutive levels are linked by a lift map, which we record here because the density series (Corollary 3.7), the density limit (Section 4), and the lower bound of Section 8 all rest on it.

Definition 2.4 (Lift, atom). Let $r \in \mathcal{O}_L^{(a,b)}$. The two residues r and $r + 2^L$ in $\{1, \dots, 2^{L+1}\}$ are the *level- $(L+1)$ lifts* of r . An obstruction $r \in \mathcal{O}_L^{(a,b)}$ is *atomic* if it is not a lift of any level- $(L-1)$ obstruction, i.e. $r \bmod 2^{L-1} \notin \mathcal{O}_{L-1}^{(a,b)}$; the set of atomic obstructions at level L is denoted $\mathcal{A}_L^{(a,b)}$, and $\mathcal{A}_L^{(a,b),G_0}, \mathcal{A}_L^{(a,b),G \neq 0}$ are its shift-zero and nonzero-shift parts.

Theorem 2.5 (Lift theorem). *For every $r \in \mathcal{O}_L^{(a,b)}$ both level- $(L+1)$ lifts r and $r + 2^L$ lie in $\mathcal{O}_{L+1}^{(a,b)}$, and conversely every non-atomic obstruction at level $L+1$ is a lift of a unique obstruction at level L . Consequently, with $h_L := |\mathcal{O}_L^{(a,b)}|$,*

$$h_{L+1} = 2h_L + |\mathcal{A}_{L+1}^{(a,b)}|.$$

Proof. For $(a, b) = (3, -1)$ this is the constructive lift of [8, Theorems 3.1 and 3.2]. We recall the mechanism and its a -dependence. The lift $r_+ := r$ runs the same reduction with the modulus doubled, so its data agree with those of r at level L until the level- $(L+1)$ stop. By the terminal synchronisation (Remark 2.3) the two tracks share an exact/strict terminal type, and the two cases are: if the terminal step at L was *exact*, one extra step at the terminal index yields, using $v_K^{(J)} - v_I^{(J)} = -\delta_J$, $c_{J+1} = c_J \cdot 2^{v_K^{(J)} - v_I^{(J)}} = 1$ and $d_{J+1} = 0$ (a shift-zero lift); if it was *strict*, the index- J data already satisfy the level- $(L+1)$ stop and the terminal certificate is unchanged (a shift- s lift); both are obstructions. The lift $r_- := r + 2^L$ differs from r_+ by the discrepancy $\Delta_K^{(j)} = 2^{L-V_K^{(j)}} a^j$ (and $\Delta_I^{(j)} = 2^{L-v-V_I^{(j)}} a^j$); because a is odd, multiplying by a preserves the 2-adic valuation, so while $V_K^{(j)} < L$ the two summands of $a a_{K,-}^{(j)} + b = (a a_{K,+}^{(j)} + b) + a \Delta_K^{(j)}$ sit at distinct valuations and the ultrametric identity forces the per-step valuations to match; the case split on the exact/strict terminal type is identical and yields the second lift. The multiplier a enters only through $c_J = 2^{\delta_J}$, $d_J = b(c_J - 1)/a$ (replacing the 4^s and $(1 - 4^s)/3$ of the $a = 3$ case) and the factor a^j in the discrepancy — none of which affects the lift count of two. The argument therefore holds for every odd a with $|a| \geq 3$ and odd $b \neq 0$. For the converse and the balance: an obstruction $r \in \mathcal{O}_{L+1}^{(a,b)}$ is either atomic — contributing to $|\mathcal{A}_{L+1}^{(a,b)}|$ — or, by Definition 2.4, satisfies $r \bmod 2^L \in \mathcal{O}_L^{(a,b)}$, in which case it is one of the two lifts of its unique parent $r \bmod 2^L$; the two alternatives are mutually exclusive and exhaustive, so $h_{L+1} = 2h_L + |\mathcal{A}_{L+1}^{(a,b)}|$. (Direct enumeration confirms the balance for all tested (a, b) and $L \leq 16$.) $\square$

2.5. Asymptotic density. The *asymptotic obstruction density* is

$$c_W^{(a,b)} := \lim_{L \rightarrow \infty} \frac{|\mathcal{O}_L^{(a,b)}|}{2^L}$$

when this limit exists. The T_- -manuscript proves existence and positivity of $c_W = c_W^{(3,-1)}$; we extend both to the entire $(an + b)$ family in Section 3 (cardinality preservation under Ψ_b) and Section 4 (existence of the limit via the lift theorem).

3. FOUNDATIONAL TOOLS

3.1. The conjugation theorem.

Theorem 3.1 (Conjugation theorem). *Let a be an odd integer with $|a| \geq 3$ and b a nonzero odd integer. For every level $L \geq 1$ the multiplicative map*

$$\Psi_b : \mathbb{Z}/2^L\mathbb{Z} \rightarrow \mathbb{Z}/2^L\mathbb{Z}, \quad \Psi_b(r) := br \bmod 2^L,$$

restricts to a bijection $\mathcal{O}_L^{(a,1)} \rightarrow \mathcal{O}_L^{(a,b)}$ that preserves the shift index: $\delta_J^{(a,b)}(\Psi_b(r)) = \delta_J^{(a,1)}(r)$ (trivially so at levels below $L_{\min}(a)$, where both sets are empty).

Proof. Since b is odd, Ψ_b is a bijection of $\mathbb{Z}/2^L\mathbb{Z}$. We must show it restricts to obstructions and preserves the shift index.

Step 1 (initial data). For $r \in \mathcal{O}_L^{(a,1)}$, $v_2(r+1) = v$ and $m_K := (r+1)/2^v$ is odd. For $r' := \Psi_b(r) = br$,

$$r' + b = br + b = b(r+1), \quad v_2(r' + b) = v_2(b) + v_2(r+1) = 0 + v = v,$$

since b is odd. So $v_K'^{(0)} := v_2(r' + b)$ equals v , and $m'_K := (r' + b)/2^v = b \cdot m_K$. The initial affine data satisfy $c_0^{(a,b)} = 2^{-v} = c_0^{(a,1)}$ and $d_0^{(a,b)} = b \cdot 2^{-v} = b \cdot d_0^{(a,1)}$.

Step 2 (step-by-step conjugation). We claim by induction:

$$a_K^{(a,b),(j)} \equiv b \cdot a_K^{(a,1),(j)} \pmod{2^{L-V_K^{(j)}}}, \quad v_K^{(a,b),(j)} = v_K^{(a,1),(j)}, \quad (1)$$

and analogously for the I-track. Base case $j = 0$: $a_K^{(a,b),(0)} = br = b \cdot a_K^{(a,1),(0)}$, and we showed v_K matches at step 0. Inductive step: write $a_K^{(a,b),(j)} = b \cdot a_K^{(a,1),(j)} + 2^{L-V_K^{(j)}} \cdot \eta_j$ for some integer η_j . Then

$$A_K^{(a,b),(j)} = a(ba_K^{(a,1),(j)} + 2^{L-V_K^{(j)}} \eta_j) + b = bA_K^{(a,1),(j)} + 2^{L-V_K^{(j)}} \cdot a\eta_j.$$

Since b is odd, $v_2(bA_K^{(a,1),(j)}) = v_2(A_K^{(a,1),(j)}) = v_K^{(a,1),(j)}$; since $v_K^{(a,b),(j)} < L - V_K^{(j)}$ (non-termination), the second term has higher 2-adic valuation, so $v_K^{(a,b),(j)} = v_K^{(a,1),(j)}$ and dividing through gives $a_K^{(a,b),(j+1)} \equiv ba_K^{(a,1),(j+1)} \pmod{2^{L-V_K^{(j+1)}}}$.

Step 3 (termination and certificate). The matching valuations yield $V_K^{(a,b),(j)} = V_K^{(a,1),(j)}$ and $V_I^{(a,b),(j)} = V_I^{(a,1),(j)}$ for every j , so termination occurs at the same index J for both. From the update rule for (c_j, d_j) and the initial data of Step 1, induction gives $c_J^{(a,b)} = c_J^{(a,1)}$ and $d_J^{(a,b)} = b \cdot d_J^{(a,1)}$. Therefore

$$\Phi_J^{(a,b)} = b(1 - c_J^{(a,b)}) + a \cdot d_J^{(a,b)} = b[(1 - c_J^{(a,1)}) + a \cdot d_J^{(a,1)}] = b \cdot \Phi_J^{(a,1)},$$

and since $b \neq 0$, $\Phi_J^{(a,b)} = 0 \iff \Phi_J^{(a,1)} = 0$.

Step 4 (shift index). $\delta_J = V_K^{(J)} - V_I^{(J)} - v$ is unchanged because all summands are unchanged. $\square$

Corollary 3.2 (T_- to T_+ involution as Ψ_{-1}). *The T_- -manuscript's involution $r \mapsto 2^L - r$ inducing a bijection $\mathcal{O}_L^{(3,-1)} \leftrightarrow \mathcal{O}_L^{(3,1)}$ ([8, Section 7]) is the special case Ψ_{-1} of Theorem 3.1.*

Corollary 3.3 (b-axis universality of cardinality and density). *For every odd $b \neq 0$ and every $L \geq 1$,*

$$|\mathcal{O}_L^{(a,b)}| = |\mathcal{O}_L^{(a,1)}|.$$

In particular, $c_W^{(a,b)} = c_W^{(a)}$ depends only on a .

We have verified Theorem 3.1 numerically for $a = 3$, $L = 12$, and all odd $b \in \{-7, -5, \dots, +13\}$: the cardinalities $|\mathcal{O}_{12}^{(3,b)}| = 409$ are identical and the shift-index assignments match under Ψ_b (code: `code/python/factor_lang_check.py`).

3.2. The diophantine identity.

Theorem 3.4 (Diophantine identity). *For every $r \in \mathcal{O}_L^{(a,b)}$ with termination index J , set*

$$X_j := 2^{V_K^{(j)}} - 2^{V_I^{(j)+v}}, \quad 0 \leq j \leq J.$$

Then

$$a^{J+1} = \sum_{j=0}^J a^{J-j} \cdot X_j. \quad (2)$$

Proof. Let $D_j := 2^{V_I^{(j)+v}} \cdot d_j$. From the d -update of Lemma 2.1,

$$\begin{aligned} D_{j+1} &= 2^{V_I^{(j+1)+v}} d_{j+1} = 2^{V_I^{(j)+v} + v_I^{(j)+v}} \cdot \frac{ad_j + b(1 - c_j)}{2^{v_I^{(j)}}} \\ &= aD_j + b \cdot 2^{V_I^{(j)+v}} (1 - c_j). \end{aligned}$$

Since $c_j = 2^{\delta_j} = 2^{V_K^{(j)} - V_I^{(j)} - v}$, $2^{V_I^{(j)+v}} (1 - c_j) = 2^{V_I^{(j)+v}} - 2^{V_K^{(j)}} = -X_j$, so the recurrence simplifies to

$$D_{j+1} = aD_j - b \cdot X_j, \quad D_0 = b. \quad (3)$$

Unrolling,

$$D_{J+1} = a^{J+1} D_0 - b \sum_{j=0}^J a^{J-j} X_j = b \left(a^{J+1} - \sum_{j=0}^J a^{J-j} X_j \right).$$

On the other hand, the certificate $\Phi_J = b(1 - c_J) + ad_J = 0$ multiplied by $2^{V_I^{(J)+v}}$ gives $aD_J = b \cdot X_J$, and one step of (3) gives $D_{J+1} = aD_J - bX_J = 0$. Combining the two expressions for D_{J+1} and using $b \neq 0$, (2) follows. $\square$

Remark 3.5. The diophantine identity (2) is independent of b : it involves only a , the cumulative valuations $V_K^{(j)}$, $V_I^{(j)}$, and v . This is the structural reason behind the conjugation theorem: Ψ_b preserves the entire valuation sequence (Step 3 of the proof of Theorem 3.1), so an obstruction at $(a, 1)$ produces the same valuation tuple at (a, b) and (2) is satisfied for the conjugated residue as well.

3.3. The order lemma.

Lemma 3.6 (Order lemma). *Let $e := \text{ord}_{|a|}(2)$, the multiplicative order of 2 modulo $|a|$. For every $r \in \mathcal{O}_L^{(a,b)}$ with termination index J ,*

$$\delta_J \equiv 0 \pmod{e}.$$

Proof. Reducing (2) modulo $|a|$ leaves only the $j = J$ term:

$$0 \equiv X_J = 2^{V_K^{(J)}} - 2^{V_I^{(J)}+v} \pmod{|a|}.$$

Since $\gcd(2, |a|) = 1$, the two powers are congruent and both invertible modulo $|a|$, so $2^{V_K^{(J)} - (V_I^{(J)}+v)} = 2^{\delta_J} \equiv 1 \pmod{|a|}$. By definition of e this gives $e \mid \delta_J$ when $\delta_J \geq 0$; when $\delta_J < 0$ the same congruence reads $2^{-\delta_J} \equiv 1 \pmod{|a|}$, so $e \mid (-\delta_J)$, hence $e \mid \delta_J$ in either case. $\square$

For $a = 3$, $e = \text{ord}_3(2) = 2$, and Lemma 3.6 specialises to the T_- -manuscript's parity lemma ([8, Lemma 2.4]): δ_J is even. The *shift- s normal form* of the T_- -manuscript becomes the *shift- s normal form to step e* , $\delta_J = s \cdot e$ with $s \in \mathbb{Z}$. Numerical values of e for small $|a|$ are collected in Table 1.

$ a $	$e = \text{ord}_{ a }(2)$	shift step
3	2	$\delta_J \in 2\mathbb{Z}$ (T_- parity)
5	4	$\delta_J \in 4\mathbb{Z}$
7	3	$\delta_J \in 3\mathbb{Z}$
9	6	$\delta_J \in 6\mathbb{Z}$
11	10	$\delta_J \in 10\mathbb{Z}$
13	12	$\delta_J \in 12\mathbb{Z}$
15	4	$\delta_J \in 4\mathbb{Z}$
17	8	$\delta_J \in 8\mathbb{Z}$
31	5	$\delta_J \in 5\mathbb{Z}$

TABLE 1. Multiplicative order $e = \text{ord}_{|a|}(2)$ and the induced shift-index step size, for small odd $|a|$.

3.4. Series representation of the density. The diophantine identity also produces a clean series for $c_W^{(a)}$. Let $h_L := |\mathcal{O}_L^{(a,b)}|$ and $a_L := |\mathcal{A}_L^{(a,b), G \neq 0}|$ (atomic count of nonzero-shift obstructions at level L).

Corollary 3.7 (Series representation). *By the lift theorem (Theorem 2.5), $h_{L+1} = 2h_L + a_{L+1} + \varepsilon_L$, where $\varepsilon_L := |\mathcal{A}_{L+1}^{(a,b), G_0}|$ tracks the G_0 -atomic contributions. Telescoping this balance gives*

$$c_W^{(a)} = \sum_{l \geq L_{\min}(a)} \frac{a_l}{2^l} + c_W^{(a), G_0},$$

where $c_W^{(a), G_0} := \sum_l \varepsilon_{l-1}/2^l$ is the contribution of the shift-zero atoms (not of the shift-zero obstructions, which are lifts already counted in the $2h_L$ term

and in fact carry the bulk of the density). By Theorem 6.1 the summand ε_{l-1} vanishes at every level (outside the negative-Mersenne case), so $c_W^{(a),G_0} = 0$; this recovers the T_- -manuscript's $c_W^{(3),G_0} = 0$ ([8, Theorem 6.2]) as the case $a = 3$.

The proof is a telescoping of the lift balance and is identical to the T_- -manuscript's ([8, Section 6.3]); the T_- -specific input is only the no- G_0 -atom statement, generalised to all $|a| \geq 3$ outside the negative-Mersenne case in Theorem 6.1.

3.5. Determinacy from valuation data. A fixed sequence of step valuations pins the seed to a single residue class. We isolate this as a lemma, used both by the existence criterion (Section 4) and by the atom bound (Proposition 6.3).

Lemma 3.8 (Determinacy). *Fix an integer $v \geq 1$, a termination index $J \geq 0$, and step valuations $v_K^{(0)}, \dots, v_K^{(J-1)} \geq 1$ and $v_I^{(0)}, \dots, v_I^{(J-1)} \geq 1$; write $V_K^{(j)} := \sum_{i < j} v_K^{(i)}$ and $V_I^{(j)} := \sum_{i < j} v_I^{(i)}$. The set of odd seeds r whose parallel reduction realises exactly this valuation sequence is either empty or a single residue class. Fixing the K -data alone pins r modulo $2^{V_K^{(J)}+1}$; fixing v together with the I -data pins r modulo $2^{V_I^{(J)}+v+1}$.*

Proof. By induction the K -iterate is affine in the seed r :

$$a_K^{(j)} = \frac{a^j r + \beta_j}{2^{V_K^{(j)}}}, \quad \beta_0 = 0, \quad \beta_{j+1} = a\beta_j + b 2^{V_K^{(j)}},$$

the recursion for β_j following from $a_K^{(j+1)} = (a a_K^{(j)} + b)/2^{v_K^{(j)}}$. The requirement $v_2(a a_K^{(j)} + b) = v_K^{(j)}$ reads $a a_K^{(j)} + b \equiv 2^{v_K^{(j)}} \pmod{2^{v_K^{(j)}+1}}$; clearing the denominator $2^{V_K^{(j)}}$ and substituting the affine form turns it into

$$a^{j+1} r \equiv 2^{V_K^{(j+1)}} - a\beta_j - b 2^{V_K^{(j)}} \pmod{2^{V_K^{(j+1)}+1}}.$$

Since a is odd, a^{j+1} is a unit modulo $2^{V_K^{(j+1)}+1}$, so this fixes r modulo $2^{V_K^{(j+1)}+1}$. The conjunction over $j = 0, \dots, J-1$ has nested moduli, so the set of seeds realising the entire K -sequence is empty or a single residue class modulo $2^{V_K^{(J)}+1}$. The identical argument on the I -track — whose seed $m = (r+b)/2^v$ is affine in r once $v := v_2(r+b)$ is fixed — shows that fixing v together with $v_I^{(0)}, \dots, v_I^{(J-1)}$ confines r to a single class modulo $2^{V_I^{(J)}+v+1}$. $\square$

4. DENSITY LIMIT AND EXISTENCE CRITERION

The lift theorem makes the obstruction densities converge to $c_W^{(a)}$ and turns the existence question $c_W^{(a)} > 0$ into a finite diophantine problem; a closing remark records the 2-adic Haar-measure reading of $c_W^{(a)}$.

Proposition 4.1 (Density limit). *The ratio $|\mathcal{O}_L^{(a,b)}|/2^L$ is non-decreasing in L and bounded above by $\frac{1}{2}$, so the asymptotic density*

$$c_W^{(a,b)} = \lim_{L \rightarrow \infty} \frac{|\mathcal{O}_L^{(a,b)}|}{2^L}$$

exists, and every finite ratio $|\mathcal{O}_L^{(a,b)}|/2^L$ is a rigorous lower bound for it.

Proof. By the lift theorem (Theorem 2.5), $h_{L+1} = 2h_L + |\mathcal{A}_{L+1}^{(a,b)}| \geq 2h_L$ with $h_L := |\mathcal{O}_L^{(a,b)}|$, so $h_{L+1}/2^{L+1} \geq h_L/2^L$. Only odd residues can be obstructions, so $h_L \leq 2^{L-1}$ and the ratio stays $\leq \frac{1}{2}$; a bounded monotone sequence converges. $\square$

Definition 4.2 (Realisable valuation datum). A *finite valuation datum* consists of an integer $v \geq 1$, a termination index $J \geq 0$, and step valuations $(v_K^{(j)})_{0 \leq j < J}$, $(v_I^{(j)})_{0 \leq j < J}$, all ≥ 1 . It is *realisable* if the residue class it pins by the determinacy lemma (Lemma 3.8) is non-empty and, for some level L , its odd representative r runs the parallel reduction with precisely this valuation sequence and *terminates at index J* (no stop fires before J , and a stop fires at J).

Theorem 4.3 (Existence criterion). *The following are equivalent:*

- (1) $c_W^{(a)} > 0$;
- (2) $\mathcal{O}_L^{(a,b)} \neq \emptyset$ for some level L ;
- (3) *there exists a realisable valuation datum (Definition 4.2) satisfying the diophantine identity (2).*

Proof. (1) $\Rightarrow$ (2) is immediate. (2) $\Rightarrow$ (1): if $r \in \mathcal{O}_{L_0}^{(a,b)}$, the lift theorem (Theorem 2.5) gives $|\mathcal{O}_L^{(a,b)}| \geq 2^{L-L_0}$ for every $L \geq L_0$ (both lifts of an obstruction are obstructions), so $c_W^{(a)} \geq 2^{-L_0} > 0$ by Proposition 4.1.

(2) $\Rightarrow$ (3): an obstruction $r \in \mathcal{O}_L^{(a,b)}$ runs a finite reduction terminating at some index J with $\Phi_J = 0$; its valuation datum is realisable (it is realised by r itself) and satisfies (2) by Theorem 3.4.

(3) $\Rightarrow$ (2): let a realisable datum satisfy (2), and let r be the odd representative pinned by the determinacy lemma (Lemma 3.8), so that the reduction at (r, L) has exactly the prescribed valuations and terminates at J . Then the recurrence (3) holds, and the computation in the proof of Theorem 3.4 gives $D_{J+1} = b(a^{J+1} - \sum_{j=0}^J a^{J-j} X_j) = 0$ by (2). Since $\Phi_J(r) \cdot 2^{V_I^{(J)}+v} = aD_J - bX_J = D_{J+1} = 0$ and $2^{V_I^{(J)}+v} \neq 0$, we get $\Phi_J(r) = 0$, so $r \in \mathcal{O}_L^{(a,b)}$. $\square$

The reduction (2) $\Leftrightarrow$ (3) recasts existence of obstructions — equivalently the conjecture $c_W^{(a)} > 0$ (Conjecture 5.5) — as a concrete diophantine solvability question for the integer identity (2) over realisable valuation data, a question internal to the finite residue tower rather than one about $\mathbb{Z}_2$ directly.

Remark 4.4 (A 2-adic Haar reading). The density has a clean measure-theoretic shadow, in the tradition of the Bernstein–Lagarias $3x + 1$ conjugacy [1]. Let $\mathbb{Z}_2$ be the ring of 2-adic integers, $\mathbb{Z}_2^\times$ its units, and μ_2 the normalised Haar measure ($\mu_2(\mathbb{Z}_2) = 1$, $\mu_2(\mathbb{Z}_2^\times) = \frac{1}{2}$; see Robert [7]). The cylinders $S_L := \bigsqcup_{r \in \mathcal{O}_L^{(a,b)}} (r + 2^L \mathbb{Z}_2) \subseteq \mathbb{Z}_2^\times$ satisfy $\mu_2(S_L) = |\mathcal{O}_L^{(a,b)}|/2^L$ and increase ($S_L \subseteq S_{L+1}$, since both lifts of an obstruction are obstructions), so their union $\mathcal{O}^{2\text{-ad},(a,b)} := \bigcup_L S_L = \{\xi \in \mathbb{Z}_2^\times : \xi \bmod 2^L \in \mathcal{O}_L^{(a,b)} \text{ for all large } L\}$ is open and, by continuity of μ_2 from below, $c_W^{(a,b)} = \mu_2(\mathcal{O}^{2\text{-ad},(a,b)})$. The reduction does *not* extend to a terminating procedure on $\mathbb{Z}_2^\times$ directly: at $(a, b) = (3, -1)$ the unit $\xi = 27$ is an obstruction at every level, yet its $\mathbb{Q}_2$ K-iterate $27 \rightarrow 5 \rightarrow 7 \rightarrow \dots$ never terminates — so $\mathcal{O}^{2\text{-ad},(a,b)}$ is the tail set above, not the zero-set of an infinite reduction.

5. $L_{\min}(a)$ AND THE CHARACTERISATION OF $J = 1$ ATOMS

The smallest level at which $\mathcal{O}_L^{(a,b)}$ is nonempty is denoted $L_{\min}(a)$ (independent of b by Theorem 3.1). We give a lower bound and an exact characterisation of the levels at which atoms with $J = 1$ exist.

5.1. Lower bound from the order lemma.

Theorem 5.1 (Lower bound). *For a outside the negative-Mersenne case, $L_{\min}(a) \geq e + 1$.*

Proof. By hypothesis a is outside the negative-Mersenne case (whose $L_{\min}$ is governed by the companion’s sub-class analysis, Section 10). The smallest non-empty level is attained by an atomic obstruction, and by Theorem 6.1 (a forward dependency, proved in Section 6; no result of Section 6 depends on the present bound, so there is no circularity) atomic obstructions have $\delta_J \neq 0$. By the order lemma (Lemma 3.6), $e \mid \delta_J$, hence $|\delta_J| \geq e$. At termination both cumulative valuations are bounded by the available modulus, $V_K^{(J)} \leq L - 1$ and $V_I^{(J)} + v \leq L - 1$, and $V_I^{(J)} + v \geq v \geq 1$; therefore, from $\delta_J = V_K^{(J)} - (V_I^{(J)} + v)$,

$$e \leq |\delta_J| \leq L - 1.$$

This gives $L \geq e + 1$, hence $L_{\min}(a) \geq e + 1$. The bound is not tight. For the Mersenne multipliers $a \in \{3, 7, 15, 31\}$ the first obstruction sits at $L_{\min}(a) = 2e + 1$ exactly (the $J = 1$ atom of Corollary 5.3); the non-Mersenne multipliers scatter around this value — e.g. $L_{\min}(5) = 11$ and $L_{\min}(21) = 10$ against $2e + 1 = 9$ and 13 respectively — and the honest regularity is the asymptotic $L_{\min}(a)/e \rightarrow 2$ of Conjecture 5.4. $\square$

5.2. Characterisation of $J = 1$ atoms.

Proposition 5.2 ($J = 1$ characterisation). *For a positive odd integer $a \geq 3$, the order $e := \text{ord}_{|a|}(2)$ gives $a \mid 2^e - 1$, so $k_a := (2^e - 1)/a \in \mathbb{Z}$. There exists*

an atomic obstruction at some level $L \leq 2e + 1$ with termination index $J = 1$ if and only if the diophantine equation

$$a - 1 + 2^v = 2^w k_a \quad (4)$$

admits integers $v \geq 1$ and w with $v + 1 \leq w \leq e$; the atom then has shift index $s = 1$ and lives at level $L = w + e + 1$, with $V_I^{(1)} = w - v$. The principal valuation $v = 1$ specialises (4) to $a + 1 = 2^w k_a$, i.e. to $q_a := (a + 1)/(2k_a) = 2^{w-1}$ being a power of 2 with $q_a \geq 2$; this $v = 1$ solution is the only one for every odd $3 \leq a \leq 19$. It is not the only one in general: the smallest exception is $a = 21$ ($e = 6$, $k_a = 3$), where (4) has no $v = 1$ solution ($q_a = 11/3 \notin 2^{\mathbb{Z}}$) but the $v = 2$ solution $20 + 2^2 = 2^3 \cdot 3$ realises a $J = 1$ atom at level $L = 10$ (the witness residue is b -dependent: $r = 317$ for $b = -1$, with $b = 1$ conjugate $\Psi_{-1}(317) = 707$). Negative multipliers are out of scope: the multiplier sign is not neutralisable by conjugation (cf. Appendix A.1).

Proof. At $J = 1$, with $V_K^{(0)} = V_I^{(0)} = 0$ so that $X_0 = 1 - 2^v$, the diophantine identity (2) $a^2 = aX_0 + X_1$ reads

$$a(a - 1) + a2^v = 2^{V_K^{(1)}} - 2^{V_I^{(1)}+v}. \quad (5)$$

Necessity. The left-hand side of (5) is positive (as $a \geq 3$), so $V_K^{(1)} > V_I^{(1)} + v$ and the shift $\delta_1 = V_K^{(1)} - V_I^{(1)} - v$ is strictly positive; by the order lemma (Lemma 3.6) it is a positive multiple of e . For an atom both cumulative valuations terminate at the boundary of their modulus, and since $V_K^{(1)} > V_I^{(1)} + v$ the binding one is the K-track, $V_K^{(1)} = L - 1$, so the hosting level is $L = V_K^{(1)} + 1 = V_I^{(1)} + v + \delta_1 + 1$. With $V_I^{(1)} \geq 1$ and $v \geq 1$, the target $L \leq 2e + 1$ gives $\delta_1 \leq 2e - V_I^{(1)} - v \leq 2e - 2 < 2e$; together with $e \mid \delta_1$ and $\delta_1 > 0$ this forces $\delta_1 = e$. Writing $w := V_I^{(1)} + v$ (so $v + 1 \leq w \leq e$, the upper bound from $L \leq 2e + 1$) and $V_K^{(1)} = w + e$, the right-hand side of (5) collapses to $2^{w+e} - 2^w = 2^w(2^e - 1) = 2^w k_a a$; dividing by a yields exactly (4).

Sufficiency. Conversely, suppose $v \geq 1$ and $v + 1 \leq w \leq e$ solve (4); we construct an atom explicitly (the determinacy lemma gives uniqueness of a realising seed, not its existence, so we exhibit one). Since k_a is odd and $w > v$, comparing 2-adic valuations in $a - 1 + 2^v = 2^w k_a$ forces $v_2(a - 1) = v$: a smaller or larger valuation of $a - 1$ would give $v_2(a - 1 + 2^v) < w$. Fix any odd b and let r be the odd residue determined modulo 2^{w+e+1} by

$$ar \equiv 2^{w+e} - b \pmod{2^{w+e+1}}, \quad (6)$$

which is well defined and odd as a is a unit and $2^{w+e} - b$ is odd; by construction $v_2(ar + b) = w + e$. The two remaining valuations follow. From $a(r + b) = (ar + b) + b(a - 1)$ and $v_2(a - 1) = v < w + e$,

$$v_2(r + b) = v_2(a(r + b)) = v_2(2^{w+e} + b(a - 1)) = v,$$

so the initial valuation is the prescribed v and $m := (r + b)/2^v$ is odd. Since $am + b = (a(r + b) + b2^v)/2^v$ and, by (6), $a(r + b) + b2^v \equiv 2^{w+e} + b(a - 1 + 2^v) =$

$$2^{w+e} + b2^wk_a \pmod{2^{w+e+1}},$$

$$am + b \equiv 2^{w-v}(2^e + bk_a) \pmod{2^{w+e+1-v}}, \quad v_2(am + b) = w - v,$$

the cofactor $2^e + bk_a$ being odd. Hence the parallel reduction of r at level $L := w + e + 1 \leq 2e + 1$ takes one step with $v_K^{(0)} = w + e$ and $v_I^{(0)} = w - v$, reaching $V_K^{(1)} = w + e = L - 1$, $V_I^{(1)} = w - v$; the residual K-modulus then has valuation $L - V_K^{(1)} = 1$, so the K-track stops and $J = 1$ (the I-step is admissible because $w - v < w + e + 1 - v$). These valuations satisfy (5) ($\delta_1 = e$), so the recurrence (3) gives $D_{J+1} = 0$ exactly as in the proof of the existence criterion (Theorem 4.3), i.e. $\Phi_1 = 0$ and $r \in \mathcal{O}_L^{(a,b)}$. Finally $V_K^{(1)} = L - 1$ is the atom boundary (Proposition 6.3, Step 2): at level $L - 1$ the K-track stops already at index 0, so $r \bmod 2^{L-1} \notin \mathcal{O}_{L-1}^{(a,b)}$ and r is atomic. The smallest non- $v=1$ instance is $a = 21$ ($v = 2$, $w = 3$), giving the $b = -1$ witness $r = 317$ at $L = 10$ ($b = 1$ conjugate 707; Table 2). $\square$

Corollary 5.3 (Mersenne construction). *Every positive Mersenne multiplier $a = 2^e - 1$ admits $J = 1$ atoms at levels $L \leq 2e + 1$. (The negative Mersenne multipliers $a = -(2^e - 1)$ also carry $J = 1$ atoms, but through the distinct subclass mechanism of the companion manuscript — where they arise alongside the $J = 0$ obstructions absent for positive a — and are not instances of this corollary; see Section 10.)*

Proof. For $a = 2^e - 1$, $k_a = 1$ and $q_a = (a + 1)/2 = 2^{e-1}$, a power of 2 of value ≥ 2 when $e \geq 2$. Apply Proposition 5.2. $\square$

The solutions of the criterion (4) and the smallest level realising $J = 1$ for small odd $|a|$ are collected in Table 2.

Conjecture 5.4 (Asymptotic $L_{\min}$). $L_{\min}(a) \leq 2e + O(1)$ as $|a| \rightarrow \infty$, and $L_{\min}(a)/e \rightarrow 2$.

The conjecture is consistent with all tested $|a| \in \{3, \dots, 31\}$; a rigorous proof requires either a characterisation of $J = j$ solvability for $j \geq 2$ or a Hensel-type lifting argument in $\mathbb{Z}_2$.

5.3. Existence for non-Mersenne multipliers.

Conjecture 5.5 (Existence). $c_W^{(a)} > 0$ for every odd a with $|a| \geq 3$.

By Theorem 4.3, this reduces to realisability of the diophantine identity (2) for at least one tuple. Empirically, $c_W^{(a)} > 0$ is confirmed for $|a| \in \{3, 5, 7, 9, 15, 17, 21, 31\}$. For $|a| \in \{11, 13, 19, 23, 25, 27\}$ no obstruction is detected at levels $L \leq 16$, but a deeper enumeration locates the first obstruction at progressively higher levels — $L = 18$ for $|a| = 11$, $L = 24$ for $|a| = 13$, and $L = 26$ for $|a| = 23$ — while $|a| \in \{19, 25, 27\}$ remain undetected through $L = 26$; this growing “waiting time” is consistent with a longer onset rather than vacuity. The enumeration underlying these statements (and the tables of Sections 5–8) is carried out for *positive* a together

$ a $	e	k_a	(v, w) solving (4)	$L^{J=1} = w + e + 1$
3	2	1	(1, 2)	5
5	4	3	none	— (≥ 11 empirical)
7	3	1	(1, 3)	7
9	6	7	none	— (≥ 11 empirical)
11	10	93	none	— (≥ 18 empirical)
15	4	1	(1, 4)	9
17	8	15	none	— (≥ 16 empirical)
21	6	3	(2, 3)	10
31	5	1	(1, 5)	11

TABLE 2. Solvability of the $J = 1$ criterion (4) $a - 1 + 2^v = 2^w k_a$ for small odd $|a|$. Entries marked “none” have no admissible (v, w) , so $J \geq 2$ throughout (their empirical $L_{\min}$ from direct enumeration up to $L = 18$ is shown). The Mersenne multipliers 3, 7, 15, 31 realise $v = 1$; $a = 21$ is the smallest multiplier whose only solution has $v \geq 2$ (code/python/count_obstructions_ac.py).

with the degenerate $a = -1$ (Section 9); by the conjugation theorem the cardinalities are b -independent, but the negative non-degenerate multipliers $a \leq -3$ are not separately enumerated here — the negative-Mersenne sub-family is treated in the companion manuscript (Section 10), and the structural results of Section 6 cover negative a through their proofs (with the negative-Mersenne caveat noted in the proof of Theorem 6.1).

6. STRUCTURAL RESULTS OVER a

6.1. Generalised no-shift-zero-atom theorem.

Theorem 6.1 (No shift-zero atom, generalised). *Let a be an odd integer with $|a| \geq 3$ that is not a negative Mersenne number $-(2^{v_M} - 1)$ ($v_M \geq 2$), let b be an odd nonzero integer, and $e := \text{ord}_{|a|}(2)$. For every level L ,*

$$|\mathcal{A}_L^{(a,b),G_0}| = 0;$$

equivalently, every atomic obstruction has nonzero shift index.

Proof. The strategy, adapting [8, Theorem 6.2], is to show that every shift-zero obstruction is a lift of a level- $(L - 1)$ obstruction, hence not atomic. The prior argument carries over under $3 \rightarrow a$, $-1 \rightarrow b$ with the parity step $\delta \in 2\mathbb{Z}$ replaced by $\delta \in e\mathbb{Z}$ (order $e := \text{ord}_{|a|}(2)$). The prior argument already forces $J \geq 2$, but trivially — for $a = 3$ no shift-zero certificate can vanish at $J \leq 1$ for any v ; what genuinely changes here is the *content* of that step: the general equation $2^v = 1 - a$ is solvable precisely for the negative-Mersenne multipliers, which it thereby isolates and forces us to exclude.

$J \geq 2$ is forced (a shift-zero obstruction needs at least two steps). A shift-zero obstruction has $\delta_J = 0$. At $J = 0$ this is impossible: $\delta_0 = V_K^{(0)} - V_I^{(0)} - v = -v \leq -1 \neq 0$ (note this is the correct obstruction to $J = 0$ — the certificate value Φ_0 itself can vanish for some negative a , but never with $\delta_0 = 0$). At $J = 1$ with $\delta_1 = 0$ we have $c_1 = 1$, so the certificate $\Phi_1 = ad_1$ vanishes only if $d_1 = 0$; the d -update then forces $ad_0 + b(1 - c_0) = 0$, i.e. $(a - 1)2^{-v} = -1$, i.e. $2^v = 1 - a$. For $a > 0$ the right-hand side is ≤ -2 , impossible. For $a < 0$ it requires $|a| + 1 = 2^v$, that is $|a|$ a Mersenne number; the negative-Mersenne multipliers are treated by the finer sub-class analysis of the companion manuscript (Section 10), while for negative non-Mersenne a the equation $2^v = |a| + 1$ has no solution. Hence, outside the negative-Mersenne case, no shift-zero obstruction has $J \leq 1$, so $J \geq 2$.

The projected reduction. Let $r \in \mathcal{O}_L^{(a,b),G_0}$ and set $r' := r \bmod 2^{L-1}$. The parallel reductions at (r, L) and $(r', L - 1)$ start from the same residue classes with the modulus halved; exactly as in the lift theorem (Theorem 2.5) the two K-tracks differ by $\Delta_K^{(j)} = 2^{L-1-V_K^{(j)}} a^j$ (and the I-tracks by $\Delta_I^{(j)} = 2^{L-1-v-V_I^{(j)}} a^j$; a odd keeps the 2-adic valuation). While $V_K^{(j)} < L - 1$ the ultrametric identity forces the per-step valuations, hence the affine data (c_j, d_j) , to agree between the two reductions. Since $J \geq 2$ the case split below is well defined.

Case A ($V_K^{(J)} < L - 1$). No level- $(L - 1)$ stop fires before index J : a K-stop at index $J' \leq J - 1$ would force $V_K^{(J)} \geq V_K^{(J'+1)} \geq L - 1$, and an I-stop at index $J' \leq J - 1$ would force $V_I^{(J)} \geq V_I^{(J'+1)} \geq L - 1 - v$, which under $\delta_J = 0$ (so $V_I^{(J)} = V_K^{(J)} - v$) again reads $V_K^{(J)} \geq L - 1$ — both contradicting the Case-A hypothesis. Hence the level- $(L - 1)$ reduction reaches index J with the same terminal data (c_J, d_J) , so $\Phi_J(r') = 0$ and $r' \in \mathcal{O}_{L-1}^{(a,b)}$ (in the same shift class G_0).

Case B ($V_K^{(J)} = L - 1$). Now the level- $(L - 1)$ K-stop fires one step earlier, at index $J - 1$ — and not before: for any $J' \leq J - 2$ one has $V_K^{(J'+1)} \leq V_K^{(J-1)} < V_K^{(J)} = L - 1$, and under $\delta_J = 0$ (so $V_I^{(J)} = V_K^{(J)} - v$) likewise $V_I^{(J'+1)} \leq V_I^{(J-1)} < L - 1 - v$, so no level- $(L - 1)$ stop precedes index $J - 1$. The $(r', L - 1)$ reduction therefore terminates at index $J - 1$ with data (c_{J-1}, d_{J-1}) . From the d -update (Lemma 2.1) the parent's certificate value is $d_J = (a d_{J-1} + b(1 - c_{J-1}))/2^{v_I^{(J-1)}}$, and the G_0 condition $d_J = 0$ gives $a d_{J-1} + b(1 - c_{J-1}) = 0$, i.e.

$$\Phi_{J-1}(r') = b(1 - c_{J-1}) + a d_{J-1} = 0.$$

Thus r' terminates as an obstruction at level $L - 1$, in shift class $\delta_{J-1}(r')/e$ (an integer by the order lemma, Lemma 3.6), so $r' \in \mathcal{O}_{L-1}^{(a,b)}$.

In both cases $r' \in \mathcal{O}_{L-1}^{(a,b)}$, so r — being one of the two lifts of r' — is not atomic (Theorem 2.5). Hence no shift-zero obstruction is atomic,

$|\mathcal{A}_L^{(a,b),G_0}| = 0$. The $J \geq 2$ forcing above is purely algebraic and holds at *every* level; the case split then applies at every L for which the projected level- $(L-1)$ reduction is defined, so the conclusion holds at all such L . The finitely many smallest levels are confirmed by direct enumeration, which gives $|\mathcal{A}_L^{(a,b),G_0}| = 0$ for all odd $3 \leq a \leq 31$ (and the degenerate $a = -1$) up to $L = 16$; the negative non-degenerate multipliers are covered by the argument, not the enumeration. $\square$

Corollary 6.2 (Synchronisation, generalised). *For every $r \in \mathcal{O}_L^{(a,b)}$ both tracks terminate at the same index J , and the cumulative valuations are synchronised to an integer multiple of e :*

$$V_K^{(J)} - (V_I^{(J)} + v) = \delta_J = e \cdot s, \quad s := \delta_J/e \in \mathbb{Z}.$$

The T_- -manuscript's case is $a = 3$, $e = 2$.

Proof. The common termination index and $v_I^{(J)} = v_K^{(J)} + \delta_J$ are the terminal synchronisation (Remark 2.3); the displayed equality is the definition $\delta_J = V_K^{(J)} - V_I^{(J)} - v$, and $e \mid \delta_J$ is the order lemma (Lemma 3.6). $\square$

6.2. Polynomial atom bound at fixed J .

Proposition 6.3 (Atom bound at fixed J). *For odd a with $|a| \geq 3$, odd $b \neq 0$, and every $J \geq 1$, the number of atomic $G_{\neq 0}$ -obstructions at level L with termination index J satisfies*

$$|\mathcal{A}_L^{(a,b),G_{\neq 0}}|_J \leq \binom{L-2}{J-1} + \binom{L-2}{J} = \binom{L-1}{J} = O_J(L^J),$$

where $|\cdot|_J$ counts atomic $G_{\neq 0}$ -obstructions with the given J . In particular every fixed- J stratum is polynomially bounded in L . The bound is unconditional; no per-step factor is left open.

Proof. The proof has three steps: a 2-adic determinacy lemma (a fixed valuation sequence pins down the residue), the atom boundary condition, and a count.

Step 1 (determinacy). By the determinacy lemma (Lemma 3.8), fixing the K-sequence $v_K^{(0)}, \dots, v_K^{(J-1)}$ pins r to at most one residue class modulo $2^{V_K^{(J)}+1}$, and fixing v together with $v_I^{(0)}, \dots, v_I^{(J-1)}$ pins it to at most one class modulo $2^{V_I^{(J)}+v+1}$.

Step 2 (atom boundary). At termination both cumulative valuations obey $V_K^{(J)} \leq L-1$ and $V_I^{(J)} + v \leq L-1$ (a successful step j requires $v_K^{(j)} < L - V_K^{(j)}$, resp. $v_I^{(j)} < L - v - V_I^{(j)}$, so $V_K^{(j+1)} \leq L-1$, resp. $V_I^{(j+1)} + v \leq L-1$). Suppose both were strict: $V_K^{(J)} < L-1$ and $V_I^{(J)} + v < L-1$. Then the discrepancy argument from the proof of the lift theorem (Theorem 2.5) applies to $r' := r \bmod 2^{L-1}$: the two reductions differ by $\Delta_K^{(j)} = 2^{L-1-V_K^{(j)}} a^j$ (and the I-analogue), and because a is odd and $V_K^{(j)} \leq V_K^{(J)} < L-1$ (resp.

$V_I^{(j)} + v \leq V_I^{(J)} + v < L - 1$) throughout, the ultrametric identity forces equal per-step valuations and hence equal terminal data, so $\Phi_J(r') = 0$ and $r' \in \mathcal{O}_{L-1}^{(a,b)}$. Then r , a lift of r' , is not atomic. Contrapositively, every atomic $G_{\neq 0}$ -obstruction satisfies the boundary condition

$$V_K^{(J)} = L - 1 \quad \text{or} \quad V_I^{(J)} + v = L - 1.$$

Step 3 (count). Combine Steps 1 and 2. If $V_K^{(J)} = L - 1$, the K-sequence pins r modulo $2^{V_K^{(J)}+1} = 2^L$, hence to at most one residue in $\{1, \dots, 2^L\}$; the admissible K-sequences are the compositions of $L - 1$ into J positive parts, $\binom{L-2}{J-1}$ in number. If instead $V_I^{(J)} + v = L - 1$, the datum $(v_I^{(0)}, \dots, v_I^{(J-1)}, v)$ pins r modulo $2^{V_I^{(J)}+v+1} = 2^L$, again to at most one residue; these data are the compositions of $L - 1$ into $J + 1$ positive parts, $\binom{L-2}{J}$ in number. By Step 2 every atom lies in at least one case, and by Step 1 each datum contributes at most one residue, so

$$|\mathcal{A}_L^{(a,b), G_{\neq 0}}|_J \leq \binom{L-2}{J-1} + \binom{L-2}{J} = \binom{L-1}{J}$$

by Pascal's rule. Since $\binom{L-1}{J} \leq (L-1)^J/J!$, this is $O_J(L^J)$. $\square$

Remark 6.4. The exponent J is generous: the boundary congruences couple the two tracks, and the order lemma (Lemma 3.6) and the realisability of each composition cut the admissible data far below $\binom{L-1}{J}$ (e.g. at $a = 3$ the $J = 5$ stratum peaks near 30 rather than $\binom{L-1}{5}$). What the bound establishes unconditionally is that each fixed- J stratum is polynomial in L ; the *summation* over J — where the strata overlap and J itself ranges up to $J_{\max} = L - 4$ — is the open content of the growth-exponent question of Section 7 (Conjecture 7.1, and the open problem whether $\alpha(a) = 1$).

Summed over all J , the total atom count $|\mathcal{A}_L^{(a,b), G_{\neq 0}}|$ fixes the growth exponent $\alpha(a) := \limsup_L |\mathcal{A}_L|^{1/L} \in [1, 2]$. Whether this total is genuinely polynomial — the *strong* reading $\alpha(a) = 1$ — or only sub-2 exponential ($\alpha(a) \in (1, 2)$) we leave as an *open problem* rather than a conjecture: the deeper empirical data of Section 7 lean against the polynomial reading. What we do conjecture is the strictly weaker bound $\alpha(a) < 2$ (Conjecture 7.1 in Section 7), which already suffices for the series representation of Corollary 3.7 to converge and is the principal *tractable* target.

Proposition 6.3 controls each fixed- J stratum polynomially, but the summation over J — where J ranges up to $J_{\max} = L - 4$ — would need a concentration estimate on the J -distribution that we do not have. The early hypothesis $J_{\max}(L) = O(\log L)$ is refuted by Theorem 6.5 below (Appendix A.2); the corrected picture is a bell-shaped J -distribution with mode $\approx \log_2 L + e$ and a heavy tail to $J_{\max} = L - 4$ ($|a| = 3$). For $a = 3$ the polynomial question is identical to the T_- -manuscript's open problem ([8, Section 6.4]), here extended to all $|a| \geq 3$.

6.3. Constructive extremal family at $J = L - 4$ for $a = 3$.

Theorem 6.5 ($L - 4$ extremal family). *For $a = 3$, $b = -1$, and every even $L \geq 6$, the residue $r := 2^{L-2} + 3$ is an atomic obstruction in $\mathcal{O}_L^{(3,-1)}$ with termination index $J = L - 4$ and shift index $s = 1$. (For odd L this particular residue is not an obstruction — the trace below yields $\delta_J = 1$, an odd shift index forbidden by the order lemma; the extremal index for odd L is not addressed here.)*

Proof. We exhibit the iteration trace. At $j = 0$: $r = 2^{L-2} + 3$, $3r - 1 = 3 \cdot 2^{L-2} + 8 = 2^3(3 \cdot 2^{L-5} + 1)$, so $v_K^{(0)} = 3$; $m = (r - 1)/2 = 2^{L-3} + 1$ (with $v = 1$), $3m - 1 = 3 \cdot 2^{L-3} + 2 = 2(3 \cdot 2^{L-4} + 1)$, so $v_I^{(0)} = 1$.

At $j = 1, \dots, L - 6$: $a_K^{(j)} = 3^j \cdot 2^{L-4-j} + 1$, $3a_K^{(j)} - 1 = 3^{j+1} \cdot 2^{L-4-j} + 2 = 2(3^{j+1} \cdot 2^{L-5-j} + 1)$, so $v_K^{(j)} = 1$; analogously $v_I^{(j)} = 1$.

At $j = L - 5$: $a_K^{(L-5)} = 2 \cdot 3^{L-5} + 1$, $3 \cdot a_K^{(L-5)} - 1 = 2(3^{L-4} + 1)$. We take L even, so $3^{L-4} \equiv 1 \pmod{4}$ and $v_K^{(L-5)} = 2$. (For odd L one has $v_K^{(L-5)} = 1$ instead; the cumulative valuation then yields $\delta_J = 1$, an odd shift index, so r fails the order lemma and is not an obstruction — hence the even- L restriction in the statement.)

At $j = L - 4$: $V_K^{(L-4)} = 3 + (L - 6) \cdot 1 + 2 = L - 1$, so $v_2(b_K^{(L-4)}) = L - V_K^{(L-4)} = 1$, and the K-track terminates at this step (assuming $v_K^{(L-4)} \geq 1$, which holds). On the I-track, $V_I^{(L-4)} = (L - 4) \cdot 1 = L - 4$, and the residual modulus $v_2(b_I^{(L-4)}) = L - v - V_I^{(L-4)} = L - 1 - (L - 4) = 3$, so the I-track also terminates. Both at $J = L - 4$.

Shift index: $\delta_J = V_K^{(J)} - V_I^{(J)} - v = (L - 1) - (L - 4) - 1 = 2 = 1 \cdot e$, so $s = 1$. Certificate: $\Phi_J = b(1 - c_J) + ad_J$ with $c_J = 2^2 = 4$ and d_J computable from the diophantine identity to $(1 - c_J)/3 = -1$; $\Phi_J = (-1)(1 - 4) + 3 \cdot (-1) = 3 - 3 = 0$. So $r \in \mathcal{O}_L^{(3,-1)}$.

Atomicity: run the reduction on $r \bmod 2^{L-1}$ at level $L - 1$. It agrees with the level- L trace until the K-track reaches $V_K^{(L-5)} = 3 + (L - 6) = L - 3$, where the residual modulus is $(L - 1) - (L - 3) = 2$; since $v_K^{(L-5)} = 2 \not\leq 2$, the K-track stops one step early, at index $L - 5$. There the shift index is

$$\delta_{L-5} = V_K^{(L-5)} - V_I^{(L-5)} - v = (L - 3) - (L - 5) - 1 = 1,$$

an *odd* value (here $V_I^{(L-5)} = L - 5$ from $v_I^{(j)} = 1$, $j \leq L - 6$). By the order lemma (Lemma 3.6, $e = 2$) no obstruction can have odd shift, so $\Phi_{L-5}(r \bmod 2^{L-1}) \neq 0$ and $r \bmod 2^{L-1} \notin \mathcal{O}_{L-1}^{(3,-1)}$. Hence r is atomic. $\square$

Conjecture 6.6 (Maximality of $L - 4$ for $a = 3$). *For $a = 3$, $b = -1$, and every even $L \geq 6$, $J = L - 4$ is the maximum termination index attained by any atomic obstruction in $\mathcal{O}_L^{(3,-1)}$.*

Empirically verified for even $L \in \{10, 12, 14, 16\}$ (code/python/count_obstructions_ac.py). A rigorous proof would require non-existence

statements at $J = L - 3, L - 2, L - 1$, which appear to follow from the diophantine identity reduced modulo $|a|^{J+1}$ but we do not pursue this here.

6.4. J -distribution: empirical picture. Empirical J -statistics at $a = 3$, $b = -1$:

L	$ \mathcal{A}_L $	$J_{\min}$	$J_{\max}$	$\bar{J}$
10	7	3	6	4.00
12	21	4	8	4.76
14	66	4	10	5.64
16	214	4	12	6.57
18	722	5	14	—

TABLE 3. Termination-index statistics for atomic $G_{\neq 0}$ -obstructions at $a = 3$, $b = -1$. $J_{\max} = L - 4$ matches Theorem 6.5; the mean $\bar{J}$ grows slowly (logarithmically) in L — broadly consistent with a $\log_2 L + e$ trend, though the few available points do not pin the constant down.

Across $|a| \in \{3, 5, 7\}$ at $L = 16$, the gap $L - J_{\max}(L, a)$ between L and the largest termination index of an *atomic* $G_{\neq 0}$ -obstruction (as in Table 3) depends on a (Table 4):

$ a $	e	L	$J_{\max}$	$L - J_{\max}$
3	2	16	12	4
5	4	16	6	10
7	3	16	5	11

TABLE 4. Atomic $J_{\max}$ across $|a|$ at $L = 16$, taken — as in Table 3 — over atomic $G_{\neq 0}$ -obstructions (the maximum over *all* obstructions is larger, e.g. 8 for $a = 5$ and 10 for $a = 7$). The T_- -manuscript value $L - J_{\max} = 4 = 2e$ at $a = 3$ does not generalise as $2e$; see Conjecture 6.7.

Conjecture 6.7 ($L - J_{\max}$ scaling). $L - J_{\max}(L, a) \approx \text{const}(a) \cdot e + O(1)$ with $\text{const}(3) = 2$.

7. THE GROWTH EXPONENT $\alpha(a)$

7.1. $\alpha(a) < 2$. Let $\alpha(a) := \limsup_L |\mathcal{A}_L^{(a), G_{\neq 0}}|^{1/L}$. The series convergence in Corollary 3.7 gives $\alpha(a) \leq 2$ trivially.

Conjecture 7.1 ($\alpha(a) < 2$ strict). *For every odd a with $|a| \geq 3$, $\alpha(a) < 2$.*

This is the weak form of the growth-exponent question ($\alpha(a) = 1 \Rightarrow \alpha(a) < 2$) and is equivalent to $|\mathcal{A}_L|/2^L \rightarrow 0$. Direct enumeration constrains $\alpha(3)$ without determining it: over $a = 3$ (Table 3 with $|\mathcal{A}_L| = 7, 21, 66, 214, 722$ at $L = 10, 12, \dots, 18$, extended to $|\mathcal{A}_{24}| = 32051$), the raw root $|\mathcal{A}_L|^{1/L}$ rises $1.44 \rightarrow 1.54$ ($L = 18, 24$) and the per-step ratio $|\mathcal{A}_L|/|\mathcal{A}_{L-1}|$ rises $1.85 \rightarrow 1.91$, with no sign of the turnover that the polynomial regime $\alpha(3) = 1$ would eventually require (a polynomial L^d has its ratio *decreasing* to 1). The data are therefore more readily read as $\alpha(3) \in (1, 2)$ than as $\alpha(3) = 1$, though a finite enumeration cannot decide a lim sup; this is why we state only $\alpha(a) < 2$ as a conjecture and leave $\alpha(a) = 1$ as an open problem.

7.2. A possible spectral route. One natural attack on Conjecture 7.1 adapts Wirsching’s Markov operator for $T_+ = (3n + 1)/2^v$ ([10]), an $L^1(\mathbb{Z}_2)$ -contraction with a unique invariant density. The analogue here would act on the parallel-reduction state (c, d) through the update rule of Lemma 2.1, weighting the transition with step valuations (v_K, v_I) by $2^{-v_K - v_I}$; a spectral gap $\lambda_1 < 1$ of such an operator would give $\alpha(a) = 2\lambda_1 < 2$. We record this only as a possible route and do not develop it: unlike Wirsching’s setting, the state space $(c, d) \in \mathbb{Q}_2 \times \mathbb{Q}_2$ is unbounded, and neither the existence of an invariant density nor a spectral gap is clear.

8. EMPIRICAL DENSITY SCALING

We report the empirical scaling of $c_W^{(a)}$ with $|a|$ via direct enumeration at level $L \leq 16$ (`code/python/count_obstructions_ac.py`). The density estimates in Table 5 are restricted to the multipliers $|a| \in \{3, 5, 7, 15, 31\}$ that are well resolved at $L = 16$. The remaining tested multipliers do carry obstructions at $L = 16$ ($|\mathcal{O}_{16}^{(9,-1)}| = 47$, $|\mathcal{O}_{16}^{(21,-1)}| = 69$, $|\mathcal{O}_{16}^{(17,-1)}| = 1$), confirming $c_W^{(a)} > 0$, but with too few obstructions for a stable density estimate at this level (for $|a| = 17$ a single obstruction); they are therefore omitted from the scaling fit.

The structural mechanism behind a 4^{-e} lower bound is the lift theorem (Theorem 2.5): a single atomic obstruction at level $L_{\min}$ produces $2^{L-L_{\min}}$ obstructions at every level $L \geq L_{\min}$, so $c_W^{(a)} \gtrsim 2^{-L_{\min}(a)}$. Granting the (conjectural) $L_{\min}(a) \approx 2e$ of Conjecture 5.4, this gives

$$c_W^{(a)} \gtrsim 2^{-L_{\min}(a)} \approx 4^{-e}.$$

Note that this argument bounds $c_W^{(a)}$ only from below; the matching upper bound is the open content of Conjecture 8.1.

Conjecture 8.1 (Bounded correction factor). *For positive odd $a \geq 3$ there exist constants $C_0 > 0$, $C_1 < \infty$ with $C_0 \leq C(a) \leq C_1$.*

The restriction to positive a is essential, not cosmetic. For the negative Mersenne multipliers $a = -(2^{v_M} - 1)$ the density scales as $c_W^{(a)} = \Theta(2^{-e})$ (Section 10), the square root of the positive scale, so $C(a) = c_W^{(a)}/4^{-e} =$

$ a $	e	4^{-e}	$c_W^{(a)}$ (empir. lower bound, $L = 16$)	$C(a) := c_W^{(a)}/4^{-e}$
3	2	0.0625	0.1153	1.84
7	3	0.0156	0.0146	0.93
15	4	0.00391	0.00243	0.62
5	4	0.00391	0.00172	0.44
31	5	0.000977	0.00053	0.55

TABLE 5. Empirical density scaling over the five multipliers well resolved at $L = 16$. Each $c_W^{(a)}$ entry is a rigorous lower bound for the limit (the sequence $|\mathcal{O}_L^{(a,b)}|/2^L$ is non-decreasing, by the lift theorem; see Proposition 4.1). A log-linear least-squares fit gives $\log_2 c_W^{(a)} \approx -2.64e + 1.91$ ($R^2 = 0.99$). The fitted slope -2.64 is somewhat steeper than the -2 that exact 4^{-e} scaling would produce: over these five points the data are consistent both with a bounded correction factor $C(a) := c_W^{(a)}/4^{-e}$ (which stays in $[0.44, 1.84]$ here) and with a slowly *decaying* $C(a)$, i.e. a decay faster than 4^{-e} . We therefore report $c_W^{(a)} \approx C(a) 4^{-e}$ as an empirical law and leave the asymptotic boundedness of $C(a)$ — the two-sided form $c_W^{(a)} = \Theta(4^{-e})$ — as Conjecture 8.1.

$\Theta(2^e) \rightarrow \infty$; the bound therefore cannot hold across the sign axis. Already the smallest negative non-Mersenne multiplier $a = -5$ ($e = 4$) leaves the positive- a band: the b -independent cardinality $|\mathcal{O}_{14}^{(-5,b)}| = 357$ gives the rigorous lower bound $C(-5) \geq 357/64 > 5.5$, against the $[0.44, 1.84]$ of Table 5. The empirical law $c_W^{(a)} \approx C(a) 4^{-e}$ and its two-sided form $c_W^{(a)} = \Theta(4^{-e})$ are thus phenomena of the *positive* multipliers; the negative multipliers are governed by the companion's sub-class analysis (Section 10) for the Mersenne case and are left uncharacterised otherwise.

Remark 8.2 (Deeper enumeration and extrapolation). The $C(a)$ entries of Table 5 are rigorous lower bounds at the truncation level $L = 16$; they understate the limit, and the gap is already visible a few levels deeper. Pushing the (unhacked) enumeration to $L = 24$ — still a rigorous lower bound by Proposition 4.1 — raises $C(3)$ from 1.84 to 2.15, and over the eight multipliers $|a| \in \{3, 5, 7, 9, 15, 17, 21, 31\}$ the lower-bound band widens to $[0.56, 4.40]$ (extremes at $a = 31$ and $a = 21$), already beyond the $[0.44, 1.84]$ read off at $L = 16$. Fitting the atomic series $|\mathcal{A}_l^{(a), G \neq 0}| \approx K_a \alpha(a)^l$ on the deepest available window and summing the geometric tail $\sum_{l > 24} |\mathcal{A}_l|/2^l$ yields *heuristic* limit estimates that stay in a comparable bounded band (e.g. $C(3) \approx 2.6$, $C(21) \approx 4.4$), with fitted growth $\alpha(a) < 2$ throughout ($\alpha(a) \in [1.46, 1.88]$ for the well-resolved $a \leq 15$; the sparser high-order multipliers are consistent but under-resolved at $L = 24$). Only the finite

lower bounds are rigorous; the extrapolation is heuristic. These deeper data leave the rigorous content untouched, but they show that the bounded- $C(a)$ phenomenon of Conjecture 8.1 is robust to the $L = 16$ truncation — the band is wider but stays finite — and they give direct empirical support to $\alpha(a) < 2$ (Conjecture 7.1) across every tested multiplier.

9. DEGENERATE EDGE CASE: $|a| = 1$

Theorem 9.1 ($a = +1$ vacuous). *For $a = +1$ and every odd $b \neq 0$,*

$$\mathcal{O}_L^{(1,b)} = \emptyset \quad \forall L \geq 1.$$

Proof. The diophantine identity (2) with $a = 1$ becomes $1 = \sum_{j=0}^J X_j$, with $X_j = 2^{V_K^{(j)}} - 2^{V_I^{(j)}+v}$. Since $V_K^{(0)} = 0$, the $j = 0$ term contributes $2^{V_K^{(0)}} = 1$; moving it across, the identity rearranges to

$$\sum_{j=1}^J 2^{V_K^{(j)}} = \sum_{j=0}^J 2^{V_I^{(j)}+v}.$$

Each step adds a positive valuation ($v_K^{(i)} = v_2(a_K^{(i)} + b) \geq 1$ because $a_K^{(i)}, b$ are odd), so the exponents $V_K^{(1)} < V_K^{(2)} < \dots < V_K^{(J)}$ are strictly increasing, and likewise $V_I^{(0)} + v < \dots < V_I^{(J)} + v$. Hence the left-hand side is a sum of J *distinct* powers of 2 and the right-hand side a sum of $J + 1$ distinct powers of 2. By uniqueness of binary representation, two equal positive integers have the same number of binary ones; this forces $J = J + 1$, a contradiction. Therefore no odd r satisfies $\Phi_J = 0$ for $a = 1$, and $\mathcal{O}_L^{(1,b)} = \emptyset$ for every L and every odd b . (The conclusion is confirmed by direct enumeration for all tested b and $L \leq 16$.) $\square$

Remark 9.2 (The degenerate case $a = -1$). For $a = -1$ the situation is different: the map value $an + b = -n + b$ vanishes at $n = b$, where the parallel reduction degenerates ($v_2(0) = \infty$ forces immediate termination). The resulting “obstructions” are an artefact of this degeneration, not of genuine synchronisation dynamics: direct enumeration gives

$$|\mathcal{O}_L^{(-1,b)}| = 2^{L-2} \quad (L \geq 2),$$

exactly half the odd residues (for $b = +1$ they are the residues $r \equiv 1 \pmod{4}$). Thus $a = -1$ is *not* vacuous; it is degenerate. Neither value of $|a| = 1$ represents the genuine obstruction theory, which requires $|a| \geq 3$: at $a = +1$ the family is empty, and at $a = -1$ the map leaves $\mathbb{N}_{\text{odd}}$ through the zero of $an + b$. The order lemma (Lemma 3.6) is therefore necessary but not sufficient for a non-degenerate non-empty family.

10. RELATION TO THE NEGATIVE-MERSENNE CASE

For $a = -(2^{v_M} - 1)$ Mersenne with $v_M \geq 2$ (so $|a| \in \{3, 7, 15, 31, 63, \dots\}$), the parallel reduction admits a finer structure than the general $(an + b)$ theory captures. The algebraic identity

$$-qr + b = 2^v(-qm + b) + (2^{v_M} - 2^v)b, \quad q := |a|,$$

has a vanishing correction term $(2^{v_M} - 2^v)b$ at $v = v_M$ that synchronises the principal class of obstructions. Iterating produces a sub-class hierarchy indexed by the signed shift index $j \in \mathbb{Z}$ (the A -, B -, C -, F_1 -, and G_1 -families). The density bound $c_W^{(a)} \geq 1/(2|a|)$ becomes asymptotically sharp as $v_M \rightarrow \infty$; with $|a| = 2^{v_M} - 1$ and $e = v_M$ this is $1/(2|a|) = \frac{1}{2} \cdot 4^{-e/2}(1 + o(1)) = \Theta(2^{-e})$. Note that this is the *square root* of the positive- a scale: the negative-Mersenne density scales as $\Theta(2^{-e})$, distinct from — and much larger than — the empirical $\Theta(4^{-e})$ of the positive multipliers in Table 5, consistent with the cardinality ratio $|\mathcal{O}^{(-q)}|/|\mathcal{O}^{(+q)}| \approx 2^e$ recorded in Appendix A.1. We treat this case in a separate manuscript and only note the relationship.

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USE OF AI TOOLS

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APPENDIX A. REFUTED HYPOTHESES

A.1. Refuted hypothesis: inclusion $\mathcal{O}^{(+q)} \subseteq \mathcal{O}^{(-q)}$. An early conjecture (henceforth A3) asserted an injection $\mathcal{O}^{(+q)} \hookrightarrow \mathcal{O}^{(-q)}$ with complement equal to a Mersenne principal class. Concrete counterexamples (taken at $b = 1$, with $\mathcal{O}^{(\pm q)} := \mathcal{O}_L^{(\pm q, 1)}$; unlike the b -independent cardinalities, the specific witness residues depend on b — at $b = -1$ the same residues lie in $\mathcal{O}^{(-q)} \setminus \mathcal{O}^{(+q)}$ instead. The first two appear already at level $L = 10$, the third at $L = 12$, since for $|a| = 31$ the smallest non-vacuous level is $L = 11$):

$$r = 29 \in \mathcal{O}^{(+3)} \setminus \mathcal{O}^{(-3)}, \quad r = 9 \in \mathcal{O}^{(+7)} \setminus \mathcal{O}^{(-7)}, \quad r = 33 \in \mathcal{O}^{(+31)} \setminus \mathcal{O}^{(-31)}.$$

The empirical cardinality ratio is $|\mathcal{O}^{(-q)}|/|\mathcal{O}^{(+q)}| \approx 2^e$ for small e and approaches 2 as $L \rightarrow \infty$, but with positive and negative obstructions structurally distinct. The lesson: the sign of the multiplier is not neutralisable via residue conjugation. A multiplicative conjugation across the sign axis exists only as a partial bijection (revised hypothesis A3'); concrete candidates such as $r \mapsto -r \bmod 2^L$ or $r \mapsto br^{-1} \bmod 2^L$ remain empirically open.

A.2. Refuted hypothesis: $J_{\max}(L) = O(\log L)$ **for atoms.** An earlier hypothesis based on the empirical mean $\bar{J} \sim \log_2 L + e$ asserted that $J_{\max}(L) = O(\log L)$ as well. This is refuted by Theorem 6.5: $J_{\max}(L, 3) = L - 4$ grows linearly. The corrected picture is a bell-shaped distribution with logarithmic mode and linear maximum — a long-tail distribution. For the open problem whether $\alpha(a) = 1$, it is the *concentration* of the distribution around its mode that matters, not a hard bound on $J_{\max}$.

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